

SHADOW RUN

Endless Nightmare

Game Design Document

Version 1.0

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Genre	Horror Endless Runner / Action
Platform	Desktop (Windows, macOS, Linux)
Engine	Python / Pygame
Target Audience	Teen to Adult (13+)

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1 Game Overview

1.1 Game Title

SHADOW RUN: Endless Nightmare

1.2 Genre

Horror-themed Endless Runner with Action and Shooter elements. The game combines the fast-paced, reflex-driven gameplay of classic endless runners with dark horror aesthetics, strategic shooting mechanics, and a unique gravity-flip system.

1.3 Platform(s)

- **Primary:** Desktop (Windows, macOS, Linux)
 - **Display:** 1280×720 native resolution, scalable
 - **Input:** Keyboard and Mouse
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1.4 Target Audience

Demographics

- **Age Range:** 13+ (Teen to Adult)
- **Player Type:** Casual to mid-core gamers who enjoy arcade-style games with high replayability
- **Interests:** Horror themes, fast-paced action, high-score chasing, reflex-based gameplay

The game appeals to players who enjoy titles such as *Canabalt*, *Jetpack Joyride*, *Limbo*, and *Dead Cells*—combining accessibility with depth and a distinct visual atmosphere.

1.5 Core Concept & Unique Hook

Elevator Pitch

"Run through an endless nightmare, flipping between ground and ceiling to survive waves of horrific creatures in a dark, atmospheric world where every second counts."

Key Differentiators

- **Gravity Flip Mechanic:** Players can instantly flip between the ground and ceiling, opening unique traversal and combat strategies
 - **Strategic Shooting:** Limited ammunition with regeneration forces players to choose engagements carefully
 - **Invisibility System:** A charge-based stealth mechanic allows tactical evasion
 - **Progressive Horror:** Difficulty scales smoothly with score, introducing new enemy types and patterns
 - **Dark Atmospheric Design:** A cohesive horror visual identity with parallax backgrounds, fog effects, and ambient audio
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1.6 Design Pillars



Accessibility

Simple controls that are easy to learn but offer depth through mastery of timing and positioning.



Tension

Constant forward momentum and limited resources create persistent pressure and excitement.



Replayability

Procedural generation, leaderboards, and progressive difficulty ensure every run feels fresh.



Atmosphere

Dark horror aesthetics and ambient audio create an immersive nightmare experience.

2.1 Endless Runner Structure

Shadow Run follows the endless runner paradigm where the player character automatically moves forward through a procedurally generated world. The game continues indefinitely until the player's health reaches zero.

Core Loop

1. **Run:** The player automatically moves forward at an increasing pace
 2. **React:** Obstacles and enemies approach; player must respond
 3. **Survive:** Use jumping, shooting, and gravity-flipping to stay alive
 4. **Score:** Distance traveled and enemies defeated contribute to score
 5. **Repeat:** Death leads to score submission and restart option
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2.2 Automatic Forward Movement

Parameter	Value	Description
Base Auto-Run Speed	4 units/frame	Initial horizontal movement speed
Base Game Speed	1.0×	Starting multiplier for all game systems
Maximum Game Speed	2.0×	Cap on speed scaling to maintain playability
Speed Curve Steepness	0.0003	Controls how gradually speed increases

Game speed increases smoothly based on the player's score, creating natural difficulty progression without jarring transitions.

2.3 Jumping Mechanics

Jump System

- **Jump Power:** -16 units (upward velocity)
- **Gravity:** 0.8 units/frame²
- **Maximum Fall Speed:** 18 units/frame
- **Ground Detection:** Collision-based with platform surfaces

Jumping is the primary defensive action, allowing players to avoid ground-based enemies and hazards. The jump arc is designed to feel responsive yet weighty, giving players precise control over their trajectory.

2.4 Shooting Mechanics

Projectile System

Bullet Speed	18 units/frame
Bullet Damage	50 HP
Bullet Size	20 pixels
Cooldown	250ms between shots
Bullet Lifetime	2000ms
Maximum Ammo	5 bullets
Ammo Regeneration	1 bullet per 10 seconds

The limited ammunition system forces strategic decision-making. Players must evaluate threats and choose when to engage versus when to evade, adding a layer of resource management to the reflex-based gameplay.

2.5 Gravity Flip Mechanic

Signature Feature

The gravity flip mechanic is the defining feature that sets Shadow Run apart from traditional endless runners.

How It Works

1. Player activates gravity flip via input
2. Gravity direction inverts (ground ↔ ceiling)
3. Player character rotates 180° during transition
4. Player lands on the opposite surface

Flip Transition Speed	25 units
Flip Cooldown	100ms minimum between flips
Transition Duration	15 frames animation
Ceiling Height	100 pixels (mirrors ground)

Strategic Applications

- Evade ground enemies by flipping to ceiling
 - Access ceiling-only paths and collectibles
 - Engage ceiling enemies on their terrain
 - Create vertical distance from projectiles
 - Chain flips for advanced traversal patterns
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2.6 Risk vs. Reward Philosophy

Design Intent

Every player action involves weighing potential gains against dangers:

- **Shooting:** Eliminates threats but depletes limited ammo
- **Gravity Flip:** Evades current threats but may introduce new ones
- **Collecting Coins:** Increases score but may require risky positioning
- **Using Invisibility:** Provides safety but has long charge time

3 Controls

3.1 Input Mapping

Action	Primary Input	Alternative Inputs
Jump	SPACE	W, UP Arrow
Gravity Flip	SHIFT	S, DOWN Arrow, F
Shoot	Left Mouse Click	X, Z, CTRL
Invisibility	Click HUD Button	—
Pause / Resume	ESC	P

3.2 Control Philosophy

Design Goals

- Minimal button count (3 core actions)
- Immediate response to input
- Multiple input options for accessibility
- Left-hand keyboard, right-hand mouse layout

Accessibility Considerations

- Both hands can perform all actions
 - No simultaneous button presses required
 - Clear visual feedback for all inputs
 - Customizable controls (future)
-

3.3 Input Assumptions

The game is designed for desktop play with standard keyboard and mouse. Controller support is not currently implemented but is planned for future updates.

Recommended Setup

- **Left Hand:** Keyboard (SPACE for jump, SHIFT for flip)
- **Right Hand:** Mouse (aim and click to shoot)

4 Enemies

4.1 Enemy Overview

Enemies in Shadow Run are designed to create varied challenges that test different player skills. Each enemy type has distinct visual design, behavior patterns, and threat levels.

Base Stats	Value
Enemy Size	100×100 pixels
Base Speed	2 units/frame
Base Damage	20 HP
Spawn Distance	800 pixels ahead of player

4.2 Ground-Based Enemies

● Basic Enemy (Slime)

Simple patrol enemy that moves back and forth within a set range.

Health	1 HP (one-shot)
Damage	20 HP
Behavior	Horizontal patrol
Patrol Range	100 pixels
Unlock Score	0 (available from start)

● Chaser Enemy

Aggressive enemy that detects the player and pursues at high speed.

Health	1 HP
Damage	30 HP (1.5× base)
Detection Range	300 pixels
Chase Speed	4 units/frame (2× base)
Unlock Score	500 points

● Shooter Enemy (Turret)

Stationary enemy that fires projectiles at the player.

Health	1 HP
Projectile Damage	10 HP
Detection Range	400 pixels
Fire Rate	1 shot per 2 seconds
Projectile Speed	6 units/frame
Unlock Score	1500 points

● Berserker Enemy

Large, dangerous enemy that charges at the player after a brief wind-up.

Health	3 HP (requires multiple hits)
Damage	40 HP (2× base)
Detection Range	500 pixels
Charge Speed	8 units/frame (4× base)
Wind-up Time	30 frames (~0.5 seconds)
Charge Duration	90 frames (~1.5 seconds)
Unlock Score	3000 points

Ghost Enemy

Ethereal enemy that phases in and out of visibility. Immune to damage when phased.

Health	1 HP (when visible)
Damage	24 HP (1.2× base)
Speed	1.6 units/frame (0.8× base)
Phase Cycle	Continuous sine wave
Immunity Threshold	Alpha < 100

4.3 Ceiling-Based Enemies

Ceiling enemies mirror ground enemy types but are positioned on the ceiling, creating threats when the player uses gravity flip.

- **Ceiling Basic:** Patrol enemy attached to ceiling
- **Ceiling Chaser:** Pursues player along ceiling surface
- **Ceiling Shooter:** Inverted turret firing downward
- **Ceiling Ghost:** Phasing enemy on ceiling

Design Note

Ceiling enemies use a purple-tinted color palette (80, 40, 120) to visually distinguish them from ground enemies.

4.4 Flying Enemies

Flying Enemy

Enemy that moves through the air, threatening both ground and ceiling positions.

Color	Green-tinted (60, 100, 80)
Speed	3 units/frame (1.5× base)
Movement	Sine-wave vertical oscillation

Spider Enemy

Drops from ceiling or rises from ground when player approaches.

Teleporter Enemy

Blinks between positions, creating unpredictable threats.

4.5 Enemy Behavior Patterns

Individual Behaviors

- **Patrol:** Back-and-forth movement
- **Chase:** Pursue player when detected
- **Shoot:** Fire projectiles at player
- **Charge:** Wind-up then rush attack
- **Phase:** Cycle visibility/immunity

Group Patterns

- **Swarm:** 3-6 enemies spawned together
- **Swarm Spacing:** 120 pixels apart
- **Swarm Unlock:** 5000+ score
- **Swarm Chance:** 0% → 25% (scales)

4.6 Difficulty Progression

Enemy spawning and variety scales smoothly with player score:

Score Range	Enemy Types	Weights (Basic/Chaser/Shooter/Berserker)
0 – 500	Basic only	100% / 0% / 0% / 0%
500 – 1500	Basic, Chaser	60% / 40% / 0% / 0%
1500 – 3000	Basic, Chaser, Shooter	40% / 35% / 25% / 0%
3000 – 5000	All types	25% / 30% / 25% / 20%
5000+	All types + Swarms	15% / 25% / 30% / 30%

Spawn Rate Scaling

Base Spawn Interval	500 pixels
Minimum Spawn Interval	150 pixels (at max difficulty)
Base Spawn Chance	40%
Maximum Spawn Chance	85%
Enemy Speed Multiplier	1.0× → 2.5×

5 Game Systems

5.1 Health System

Maximum Health	100 HP
Starting Health	100 HP
Invincibility on Hit	1500ms (1.5 seconds)
Health Pack Restore	25 HP

Damage Sources

- Contact with basic enemies: 20 HP
- Contact with chasers: 30 HP
- Contact with berserkers: 40 HP
- Enemy projectiles: 10 HP

Invincibility Frames

After taking damage, the player becomes temporarily invincible and blinks to indicate this state. This prevents chain-damage scenarios and gives players recovery time.

5.2 Ammo System

Ammunition Management

Maximum Ammunition	5 bullets
Starting Ammunition	5 bullets
Regeneration Rate	1 bullet per 10 seconds
Regeneration Type	Passive (always regenerating)

The limited ammo pool with slow regeneration creates meaningful resource decisions. Players cannot rely solely on shooting and must incorporate jumping and gravity-flipping into their survival strategy.

5.3 Invisibility System

Stealth Mechanic

Invisibility Duration	3 seconds when activated
Charge Time	15 seconds to fully charge
Player Alpha (Invisible)	80 / 255 (semi-transparent)

While invisible, enemies cannot detect or damage the player. This provides an emergency escape option but requires careful timing due to the long charge period.

5.4 Score System

Score Sources

- **Distance:** Points accumulated based on distance traveled
- **Coins:** 10 points per coin collected
- **Enemies:** Points for eliminating enemies
- **Survival Time:** Bonus points for staying alive

Difficulty Milestones

Every 1000 points triggers a difficulty milestone notification, providing feedback on progression without disrupting gameplay.

5.5 High Score Persistence

Leaderboard System

- **Storage:** JSON file (data/leaderboard.json)
 - **Maximum Entries:** 10 scores
 - **Data Stored:** Player name, score, date
 - **Display:** Accessible from main menu
-

5.6 Difficulty Scaling System

The difficulty system uses smooth mathematical curves rather than discrete levels, creating imperceptible increases that feel natural.

Parameter	Scaling Method	Range
Game Speed	Ease-out quadratic	1.0× → 2.0×
Spawn Interval	Ease-in quadratic	500 → 150 pixels
Spawn Chance	Linear interpolation	40% → 85%
Enemy Speed	Linear interpolation	1.0× → 2.5×

Maximum Difficulty Score: 10,000 points (soft cap; continues slowly beyond)

Maximum Difficulty Multiplier: 3.0×

6 Level & World Design

6.1 Ground and Ceiling Layout

Ground Surface

- Height: 100 pixels
- Color: Dark brown (25, 20, 15)
- Top highlight: (35, 30, 25)
- Continuous solid platform

Ceiling Surface

- Height: 100 pixels (mirrors ground)
 - Same color palette as ground
 - Inverted collision detection
 - Accessible via gravity flip
-

6.2 Procedural Generation

World Generation System

The WorldGenerator class creates infinite terrain ahead of the player while cleaning up passed segments to maintain performance.

Generation Rules

- Terrain generates when camera approaches edge
 - Segments behind player are removed from memory
 - Enemy spawn points are distributed procedurally
 - Collectible placement follows weighted randomization
-

6.3 Environmental Hazards

While the primary challenges come from enemies, the environment contributes to atmosphere and spatial awareness:

- Limited visibility in fog areas
 - Visual distractions from parallax elements
 - Narrow safe zones between enemy patterns
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6.4 Atmosphere & Mood

Dark Night Setting

The game takes place in an endless nightmare landscape featuring:

- Deep night sky gradient (near-black to dark purple)
- Silhouetted dead trees and mountains
- Drifting fog particles
- Pale moon in the background
- Sparse, eerie star field

7.1 Visual Theme

Shadow Run employs a **dark horror aesthetic** inspired by games like *Limbo*, *Inside*, and *Dead Cells*. The visual style emphasizes silhouettes, high contrast, and atmospheric lighting.

Art Direction Goals

- Create oppressive, unsettling atmosphere
 - Maintain gameplay clarity despite dark palette
 - Use color strategically to highlight threats and pickups
 - Achieve cohesive, polished indie horror look
-

7.2 Color Palette

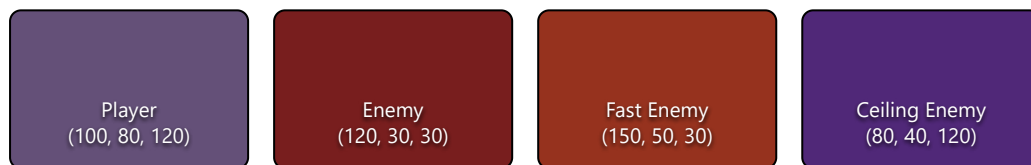
Background Colors



UI Colors



Entity Colors



Collectible & Effect Colors



7.3 Character Visibility

Contrast Rules

- **Player:** Purple-gray tone stands out against dark backgrounds
- **Enemies:** Red tones create immediate threat recognition
- **Collectibles:** Warm gold/yellow attracts attention
- **Projectiles:** Bright orange/white for tracking

All gameplay-critical elements maintain high contrast against the dark environment while decorative elements remain subtle.

7.4 Animation

Animation	Frame Count	Speed
Player Idle	4 frames	100ms/frame
Player Walk	4 frames	100ms/frame
Player Jump	4 frames	100ms/frame
Player Dash	4 frames	100ms/frame
Enemy Bob	2 frames	Sine wave

7.5 Visual Effects

- **Parallax Scrolling:** 4 background layers at different speeds
- **Fog Particles:** Drifting atmospheric particles
- **Hit Flash:** White flash on damage
- **Invincibility Blink:** Transparency oscillation
- **Berserker Charge Trail:** Motion blur effect
- **Death Particles:** Explosion on enemy death

8 Audio Design

8.1 Audio Philosophy

The audio design follows a **AAA-style dynamic mixing system** inspired by games like *Dead Space*, *Limbo*, and *Inside*. Sound reinforces the horror atmosphere while providing clear gameplay feedback.

8.2 Background Music & Ambience

Layered Ambient System

The AmbientMixer class manages multiple audio layers that crossfade based on gameplay intensity:

Layer	Description	Base Volume
Horror Ambience	Unsettling drones, distant sounds	40%
Pond Ambience	Natural, eerie outdoor sounds	35%

Dynamic Mixing

- Volume levels adjust based on gameplay intensity
- Smooth transitions (fade speed: 0.02 per frame)
- Loops seamlessly without noticeable repetition

8.3 Sound Effects

Action	Sound Description	Purpose
Jump	Whoosh / light impact	Confirm action, spatial awareness
Shoot	Sharp projectile fire	Confirm action, feedback
Hit Enemy	Impact / squelch	Damage confirmation
Take Damage	Pain / thud	Alert player to danger
Gravity Flip	Woosh / inversion tone	Orientation feedback
Collect Coin	Chime / pickup sound	Reward feedback
Invisibility Activate	Ethereal whoosh	State change indication
Enemy Death	Defeat sound / pop	Elimination confirmation

8.4 Audio Settings

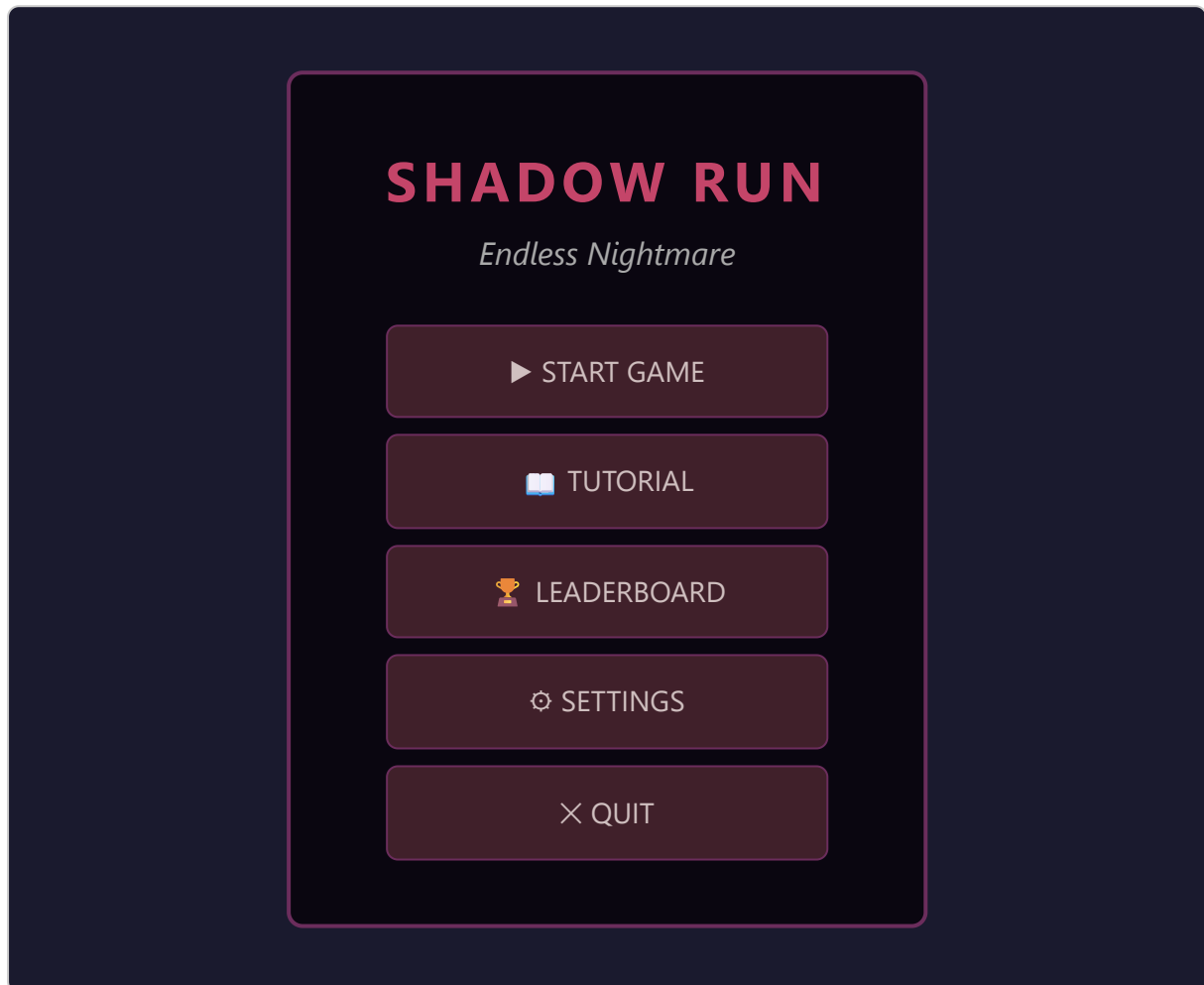
Music Volume (Default)	50%
SFX Volume (Default)	70%
Audio Channels	16 (reserved for mixing)

Procedural Sound Generation

If audio files are not present, the game generates procedural sounds programmatically to ensure audio feedback is always available.

9 UI/UX Design

9.1 Main Menu



Menu Features

- Animated background with parallax effect
 - Button hover effects with color transitions
 - Keyboard and mouse navigation
 - Horror-themed fonts and styling
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9.2 Tutorial Screen

Interactive onboarding that teaches core mechanics:

1. Movement explanation (automatic running)
 2. Jump mechanics and timing
 3. Shooting and ammo management
 4. Gravity flip introduction
 5. Invisibility system overview
 6. Enemy types preview
-

9.3 Heads-Up Display (HUD)



HUD Elements

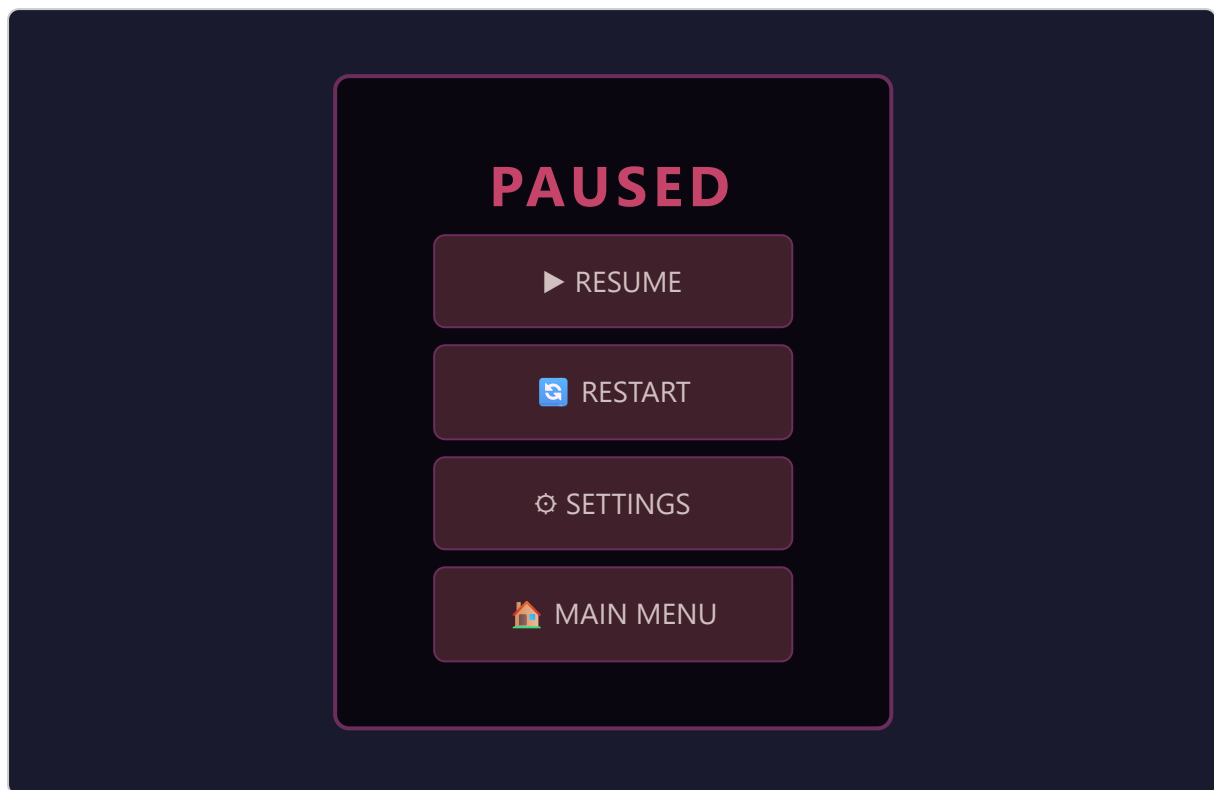
Element	Position	Information
Health Bar	Top-left	Current HP / Max HP, red fill
Ammo Bar	Below health	Current bullets / Max, orange fill
Score	Top-right	Current run score
Coins	Below score	Coins collected this run
Best Score	Below coins	Personal high score
Invisibility	Bottom-right	Charge status and activation button

Controls Hint Overlay

During the first 12 seconds of gameplay, control hints appear on screen:

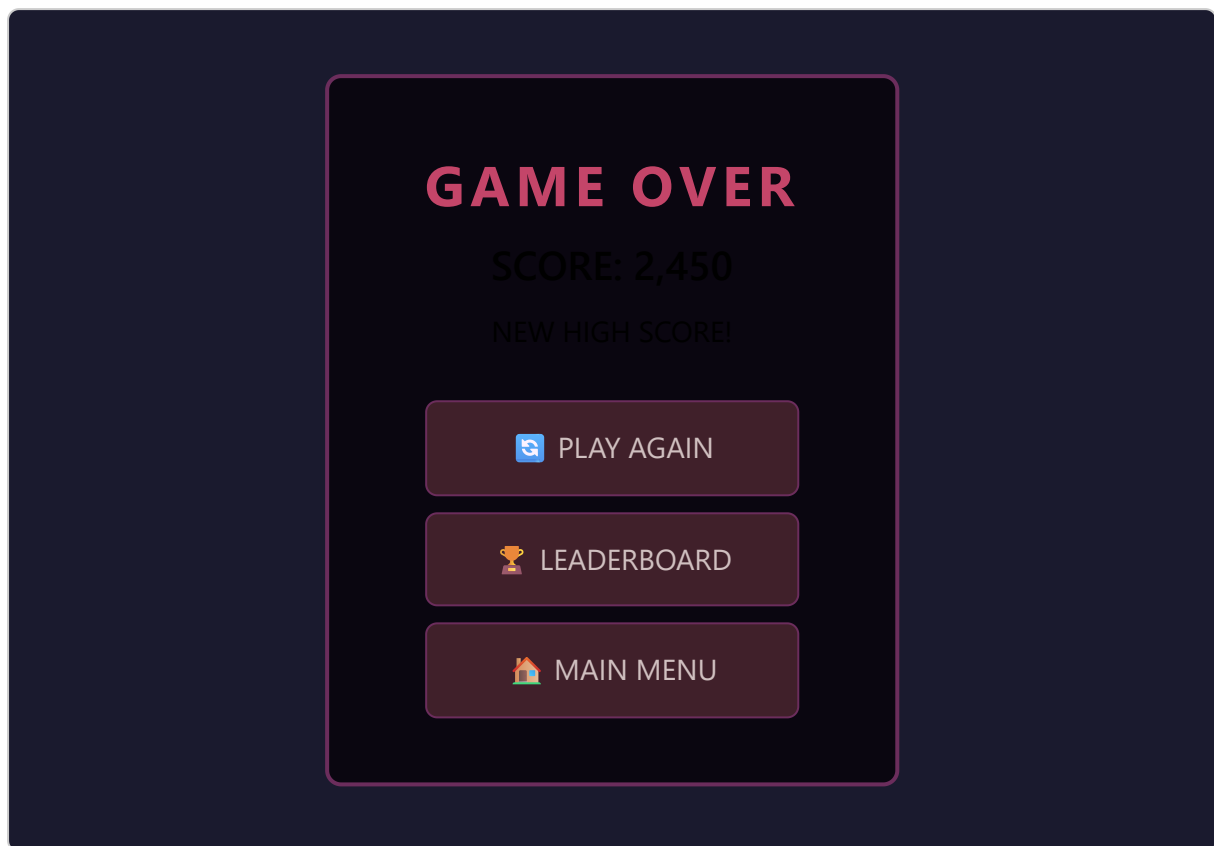
- Hints fade out starting at 7 seconds remaining
 - Shows all core actions with key bindings
 - Color-coded for quick recognition
-

9.4 Pause Menu



Semi-transparent overlay that darkens the gameplay while displaying options.

9.5 Game Over Screen



Features

- Final score display with animation
- High score notification if applicable
- Name entry for leaderboard submission
- Quick restart option

10 Technical Design

10.1 Technologies

Component	Technology	Version
Language	Python	3.8+
Game Framework	Pygame	2.x
Data Storage	JSON	—
Audio Format	OGG Vorbis	—
Image Format	PNG	—

10.2 Architecture

```
Shadow-Run/
├─ unigame.py          # Entry point
├─ game/
│  ├─ __init__.py
│  ├─ game.py          # Main game manager
│  ├─ player.py        # Player class
│  ├─ enemies.py       # Enemy classes
│  ├─ platforms.py     # Ground/ceiling generation
│  ├─ collectibles.py  # Coins, health, powerups
│  ├─ difficulty.py    # Difficulty scaling
│  ├─ camera.py        # Camera system
│  ├─ sound.py         # Audio manager
│  ├─ background.py    # Parallax backgrounds
│  ├─ ui.py            # All UI elements
│  ├─ leaderboard.py   # Score persistence
│  ├─ settings.py      # Constants & config
│  └─ utils.py         # Helper functions
├─ assets/
│  ├─ audio/           # Sound files
│  └─ sprites/         # Image assets
└─ data/
   └─ leaderboard.json
```

10.3 Resolution & Display

Native Resolution	1280 × 720 (720p)
Aspect Ratio	16:9
Target Frame Rate	60 FPS
Display Mode	Windowed

10.4 Performance Considerations

- **Object Pooling:** Reuse sprite objects to reduce allocation
 - **Culling:** Only update/render objects near camera
 - **Procedural Cleanup:** Remove off-screen world segments
 - **Cached Calculations:** Difficulty values cached until score changes
 - **Sprite Groups:** Pygame's optimized group rendering
-

10.5 Browser/Platform Compatibility

Supported Platforms

- **Windows:** 10, 11 (primary target)
- **macOS:** 10.14+ (Mojave and later)
- **Linux:** Ubuntu 18.04+, other distros with Python 3.8+

Dependencies

- Python 3.8 or higher
- Pygame 2.0 or higher
- No additional external dependencies

11.1 Additional Mechanics



Double Jump

Unlockable ability allowing a second jump in mid-air.

Priority: High



Shield Power-Up

Temporary damage absorption collectible.

Priority: Medium



Dash Ability

Quick horizontal burst for dodging and closing gaps.

Priority: Medium



Weapon Variants

Different projectile types (spread, pierce, explosive).

Priority: Low

11.2 New Enemy Types

Bat Swarm

Small, fast enemies that attack in coordinated groups.

Watcher

Stationary enemy that buffs nearby enemies when player is detected.

Phaser

Enemy that teleports between ground and ceiling.

Boss Encounters

Large-scale enemies at score milestones with unique patterns.

11.3 Visual & Audio Enhancements

- Additional parallax layers and environmental variety
 - Weather effects (rain, lightning, mist)
 - Screen shake on impacts and explosions
 - Particle system expansion (blood, sparks, debris)
 - Dynamic music intensity based on threat level
 - Voice lines or text barks for enemies
-

11.4 Platform Expansion

Platform	Feasibility	Notes
Web (HTML5)	High	Pygbag or similar for browser deployment
Mobile (iOS/Android)	Medium	Touch controls, performance optimization needed
Console	Low	Would require engine port (Unity/Godot)

11.5 Social Features

- Online leaderboards with global rankings
- Daily/weekly challenges with unique modifiers
- Replay system for sharing runs
- Achievement system with unlock rewards
- Cosmetic unlocks for player character

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